**Hashemite University**
**College of Engineering**
**Department of Computer Engineering**
**Data Science (3 Credit Hours/Dept. Compulsory)**
**(Fall Semester 2026/2027)**

| **Instructor** | | **Grading info** | | **Class Info** | |
|---|---|---|---|---|---|
| Dr.Mohammad Al-hammouri | | Project & Quizzes | 30% | Days | Sun/Tue/Thu |
| Email: | alhammouri@hu.edu.jo | Midterm Exam | 30% | Time | 11:30 - 12:30 pm |
| Office: | 3042 | Final Exam | 40% | Location | |
| Office hours will be announced soon. | | | | | |

**Course**

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|---|---|
| **Course Number:** | 2104081410 |
| **Prerequisite:** | 110408300 and 110408450<br>Students are assumed to have a good background in mathematics, particularly linear algebra and statistics. And good programming skills in Python. |
| **Textbook:** | **Text 1: "**Introduction to Python programming pdf for computer science and data science", Deitel and Deitel, 2020.<br>https://deitel.com/intro-to-python-for-computer-science-and-data-science/<br><br>**Text 2:** "Hands-on Machine Learning with Scikit-Learn, Keras, and TensorFlow" by Aurélien Géron, 2019<br><br>**Text 3: "**Python for Data Analysis", Wes McKinney, 2012.<br><br>https://www.oreilly.com/library/view/python-for-data/9781449323592/ |
| **Course Description:** | This course aims to study, application, and development of methods to learn from data. It provides an introduction to concepts, techniques, and applications in data science. It is focused on analyzing data (manipulating, processing, cleaning, and crunching data using real-world examples) in Python using different libraries such as Pandas, Matplotlib, scikit-learn, TensorFlow, and others. In addition, it provides an introduction to high-performance computing concepts to solve Big Data problems. Students then engage in case study exercises to develop and present a Big Data concept for a specific real-world case. |
| **Specific Outcomes of Instruction (Course Learning Outcomes)** | - Understand Fundamental Python Data Structures and Libraries for Data Scientists (1,2)<br>- Understand and apply the required techniques for data preprocessing visualization (2,6)<br>- Apply machine learning libraries that offer simple and efficient tools for common tasks in data analysis, such as classification, regression, clustering, dimensionality reduction, model selection, and preprocessing. (1,2,6)<br>- Discuss and Analyze real-world datasets and develop solutions and proposals for such cases. (2,6,7) |
| **Important material** | - Lecture notes<br>- Textbooks<br>- Internet resources |

Major Topics Covered and Schedule in Weeks:

| **Topic** | **# Weeks** | **# Contact hours*** |
|---|---|---|
| Introduction to Python Programming:<br><br>- Python Language Basics<br>- Built-in Data Structures, Functions, and Files | 1 | 6 |

| | | |
|---|---|---|
| - Web Integrated Development Environment (WIDE): Jupyter | | |
| Fundamental Python Libraries for Data Scientists:<br>- User Numeric and Scientific Computation: NumPy and SciPy<br>- PANDAS: Python Data Analysis Library | 1 | 6 |
| Data Preparation (Prepare the Data for Machine Learning):<br>- Data Loading, Storage, and File Formats<br>- Data Cleaning and Preparation<br>- Data Plotting and Visualization<br>- Correlation | 4 | 9 |
| Data Warehousing and On-Line Analytical Processing | 1 | 4 |
| The Fundamentals of Machine Learning:<br>- Introduction to Machine Learning<br>- Training and Testing models<br>- Supervised Learning techniques: Regression and Classification analysis<br>- Unsupervised Learning: Clustering Analysis<br>- PCA<br>- Neural Networks and Deep Learning | 5 | 9 |

**Course Policy**

- The course will follow selected subjects as listed on the course schedule.
- Students are responsible for reading extra material to enhance their knowledge in the concepts discussed during the course.
- Students are responsible for solving assignments and extra problems given in the lectures.
- Students are expected to attend and be on time. If students miss more than 15% of classes, you will automatically fail the class.

**Student Outcomes (SO) Addressed by the Course:**

| *#* | *Outcome Description* | *Contribution* |
|---|---|---|
| ***General Engineering Student Outcomes*** | | |
| (1) | An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics | ***H*** |
| (2) | An ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors | ***H*** |
| (3) | An ability to communicate effectively with a range of audiences | |
| (4) | An ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts | |
| (5) | An ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives | |
| (6) | An ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions | ***L*** |
| (7) | An ability to acquire and apply new knowledge as needed, using appropriate learning strategies | ***L*** |

**H**=High, **M**= Medium, **L**=Low

**Prepared by: Dr. Mohammad Al-hammouri** **Date: OCT, 12, 2026**